T.S.Sai Lohesi
192325137

Assignment-I

1. Mapping Data onto Aesthetics

Aim: To understand how data variables are mapped to visual properties such as color, size, position and shape

a. Using the mtcars dataset, create a scatterplot of weight (wt) vs mileage (mpg). Map the no of cyclinders to color and horsepower to point size. Add proper life, axis labels & legend titles

Code:

```
data(mtcars)
ggplot(mtcars, aes(x=wt, y=mpg, color=factor(cyl), size=hp)) +
  geom_point(alpha=0.8)+
  labs(title="Weight vs Mileage of cars",
       x="Weight(1000 lbs)",
       y="Mileage (Miles per gallon)",
       color="no. of cyclinders",
       Size="horsepower")+
theme_minimal()
```

Interpretation

The scatterplot represents the relationship blw vehicle wt. & mileage. Different colours distinguish the no. of cylinders, while point size represent horsepower

b. Examine the structure of mpg dataset. classify every variable as quantitative or qualitative and suggest a suitable asthetic for each type. Display the result as table

Code:

```
data(mpg)
Str(mpg)
```

```
variable_table <- data.frame(
    Variable_names(mpg),
    Data_type = c("Qualitative", "Qualitative", "Qualitative", "Quantitative",
                  "Quantitative", "Qualitative", "Qualitative", "Quantitative",
    Suitable_Aesthetic = c("color/shape", "label", "x/y/size", "x/y",
                           "color", "x/y/size", "color/shape", "color/shape",
                           "x/y/size", "x/y/size")
)
print(variable_table)
```

Interpretation:
The mpg dataset contains 11 variables, including both quantitative & qualitative variables. Quantitative variables represent numerical measurement, while qualitative variables represent categories

2. Coordinate Systems and Axes:

Aim: To understand cartesian and polar coordinate systems & how coordinate transformation affect visualization

a. Plot Sepal length vs Petal length from iris in cartesian coordinates with fixed aspect ratio of 1, since both variable share the same unit (cm). Add the line $y = x$ for reference

Code:

```
data(iris)
ggplot(iris, aes(x=Sepal.Length, y=Petal.length)) +
  geom_point(color="steelblue", size=2) +
  geom_abline(intercept=0, slope=1, color="red", linetype="dashed") +
  coord_fixed(ratio=1) +
  labs(title="Sepal length vs Petal length", x="Sepal length (cm)",
       y="Petal length (cm)") +
  theme_minimal()
```

Interpretation:
The scatterplot of iris flow
axes

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Interpretation:
The scatterplot shows the relationship b/w sepal length & petal length of iris flowers. The red dashed line represents equal vals on both axes

Using airquality, plot daily temp (Temp) against Day for each month in Polar coordinates. Discuss when polar coordinates are useful

Code:
```
data(airquality)
airquality$month <- factor(airquality$month)
ggplot(airquality, aes(x = Day, y = Temp, color = Month, group = Month))+
  geom_line(linewidth = 1)+
  geom_point(size = 1.5)+
  coord_polar(theta = "x")+
  labs(title = "Daily Temperature by month", x = "Day of month",
       y = "Temperature (Fahrenheit)", color = "month")+
  theme_minimal()
```

Interpretation:
The polar plot representation daily temp variations across diff months. Each month is distinguished by a different color. Polar coordinates are useful for cyclic or periodic data such as time of day, days of a cycle and seasonal patterns.

c. Create a bar chart of vehicle counts by class from mpg and convert it into a coxcomb (Polar bar) chart using coord_polar()

Code:
```
data(mpg)
ggplot(mpg, aes(x = class, fill = class))+
```

```
geom_bar()+
coord_polar(theta="y")+
labs(title="Coxcomb chart of vehicle classes", x="vehicle class",
     y="No. of vehicles", fill="vehicle class")+
theme_minimal()
```

Interpretation:

The Coxcomb chart represent vehicle counts in different classes in a circular layout. Each segment represents a vehicle class, allowing comparison of the relative no. of (which) vehicles accross categories

3. Nonlinear Axes

Aim: To understand how linear, logarithmic and square root scales help visualize data with different ranges and distributions.

a. Using the msleep dataset, plot brain wt vs body wt on linear axes & thn on log-log axes. Comment on which representation is more informative

Code:

```
data(msleep)
sleep_data <- msleep %>%
filter(!is.na(bodywt), !is.na(brainwt), bodywt>0, brainwt>0)
P1<- ggplot(sleep_data, aes(x=bodywt, y=brainwt))+
  geom_point(color="steelblue", alpha=0.7)+
  labs(title="Linear Scale", x="Body weight(kg)", y="Brain wt(kg)")+
  theme_minimal()
P2<-ggplot(sleep_data, aes(x=bodywt, y=brainwt))+
  geom_point(color="darkred", alpha=0.7)+
  Scale_x_length()+
  Scale_y_log10()+
labs(title="log-log Scale", x="Body weight(kg)-log Scale",
```

```
    y = "Brain weight (kg) - log scale") +
  theme_minimal()
P1
P2
```

Interpretation:
The linear-scale boxplot displays the original price distribution, where extreme prices can make smaller values difficult to compare. The square root scale boxplot compress large

b. Using the diamonds dataset draw boxplots of price by cut on a linear axis and on a square-root axis. Explain when a square-root scale is approximate

Code:

```
data(diamonds)
P1 <- ggplot(diamonds, aes(x = cut, y = price, fill = cut)) +
  geom_boxplot() +
  labs(title = "Diamond Price - Linear Scale", x = "Diamond cut", y = "Price") +
  theme_minimal() +
  theme(legend.position = "none")
P2 <- ggplot(diamonds, aes(x = cut, y = price, fill = cut)) +
  geom_boxplot() +
  scale_y_sqrt() +
  labs(title = "Diamond Price - Square root Scale", x = "Diamond cut",
       y = "Price (Square root Scale)") +
  theme_minimal() +
  theme(legend.position = "none")
P1
P2
```

Interpretation:
The linear-scale boxplot displays the original price distribution. Extreme prices can make smaller values difficult to compare. The sqrt scale boxplot compresses large values

4 Color Scales: Qualitative, Sequential and Diverging

Aim: To understand how qualitative, sequential and diverging color represent categorial and numerical data

a. use color as a tool to distinguish groups: plot Sepal.length vs Sepal.width of iris colored by Species using the colorblind. Safe okabe-Ito qualitative palette

Code:
```
data(iris)
okabe_ito <- c("setosa" = "#0072B2", "versicolor" = "#E69F00", "virginica
               = "#009E73")
ggplot(iris, aes(x=Sepal.length, y=Sepal.length, color=Species))+
geom_point(size=3)+
scale_color_manual(values=okabe_ito)+
labs(title="Iris Species using Okabe-Ito Palette", x="Sepal.length
     (cm)", y="Sepal width (cm)", color="Species")+
theme_minimal()
```

Interpretation:
The scatterplot shows the relationship between Sepal length and Sepal width for three iris species. The Okabe-Ito Palette distinguishable colours, making the visualization easier to interpret.

b. Use color to represent data values: draw a raster heatmap of the 2D density in faithfuld with a sequential viridis palette.

Compare quantities using horizontal
clarked box charts

Code:

```
data(faithfuld)
ggplot(faithfuld, aes(x=eruption, y=waiting, fill=density))+
geom_raster()+
scale_fill_viridis_c(option="viridis")+
labs(title="2D Density Heatmap of old Faithful Greyser",
     x="Eruption Duration", y="Waiting time", fill="Density")+
theme_minimal()
```

Interpretation:

The heatmap represents the density distribution of eruption duration and waiting time. Areas with higher density indicate combination that occur more frequently while lower-density areas indicate less common combinations.

c. Compute the correlation matrix of mtcars and display it as a heatmap using a diverging color scale centred at 0. Label each tile with its value

Code:

```
data(mtcars)
car_matrix<-cor(mtcars)
car_data<-as.data.frame(as.table(car_matrix))
ggplot(car_data, aes(x=Var1, y=Var2, fill=Freq))+
  geom_tile(color="white")+
  geom_text(aes(label=round(Freq,2)), size=2.5)+
  scale_fill_gradient2(low="blue", mid="white", high="red", midpoint
                       =0, limits=c(-1,1))+
labs(title="Correlation Matrix of mtcars", x="Variables", y="Variables
```

, fill = "Correlation") +
theme_minimal()
theme(axis.text.x = element_text(angle = 45, hjust = 1))

Interpretation:
The correlation heatmap displays the relationships blw different variables in the mtcars dataset. Blue representation negative correlations near zero and red represents positive correlations, white

5. **Color as a Tool to Highlight and Continuous Vs Discrete Scale**

Aim: To understand how color highlights important observations and distinguishes continuous and discrete variables.

a. Use Color as a tool to highlight: plot mpg vs wt. for mtcars. Showing cars with mpg > 30 in a bright accent color and all other car in grey. Label the highlighted cars.

Code:

```
data(mtcars)
car_data <- data.frame(car = rownames(mtcars), mtcars)
ggplot(car_data, aes(x = wt, y = mpg, color = mpg > 30)) +
  geom_point(size = 3) +
  geom_text(data = filter(car_data, mpg > 30), aes(label = car), vjust =
            -0.8, color = "black", size = 3) +
  scale_color_manual(values = c("FALSE" = "grey70", "TRUE" = "MPG > 30")
  ) +
  labs(title = "Highlighting Cars with Mileage Above 30", x = "Weight
       (1000 lbs)", y = "Mileage (MPG)", color = "Car category") +
  theme_minimal()
```

"sualizing Amounts:

To visualize and compare quantities using horizontal bar
arts, grouped bar charts and stacked bar charts

. draw a horizontal bar chart of the number of
each vehicle class, sorted from most to least frequent

nterpretation:

The Scatterplot highlights cars with mileage greater than 30MPG using red color, while other cars are displayed in grey. The names of highlighted cars are displayed as labels, making important

b. Using iris, plot Sepal length vs Petal length twice: once mapping Petal width to a continuous color scale and once mapping Species to a discrete color scale. Explain the difference between the two scales

Code:

```
data(iris)
P1 <- ggplot(iris, aes(x = Sepal.Length, y = Petal.Length, Color = Petal.Width)) +
geom_point(size = 3) +
scale_color_viridis_c() +
labs(title = "Continuous Color Scale", x = "Sepal length (cm)", y = "Petal
     length (cm)", Color = "Petal width") +
theme_minimal()
P1
P2 <- ggplot(iris, aes(x = Sepal.Length, y = Petal.Length, Color = Species)) +
 geom_point(size = 3) +
Scale_color_manual(values = c("Setosa" = "red", "versicolor" = "blue",
                              "viriginica" = "darkgreen")
) +
labs(title = "Discrete Color Scale", x = "Sepal length (cm)",
          y = "Petal length (cm)", Color = "Species") +
theme_minimal()
P2
```

Interpretation:

The continuous color scale represents numerical petal width with a gradual color transition. The discrete color scale represents different iris species using seperate colours.

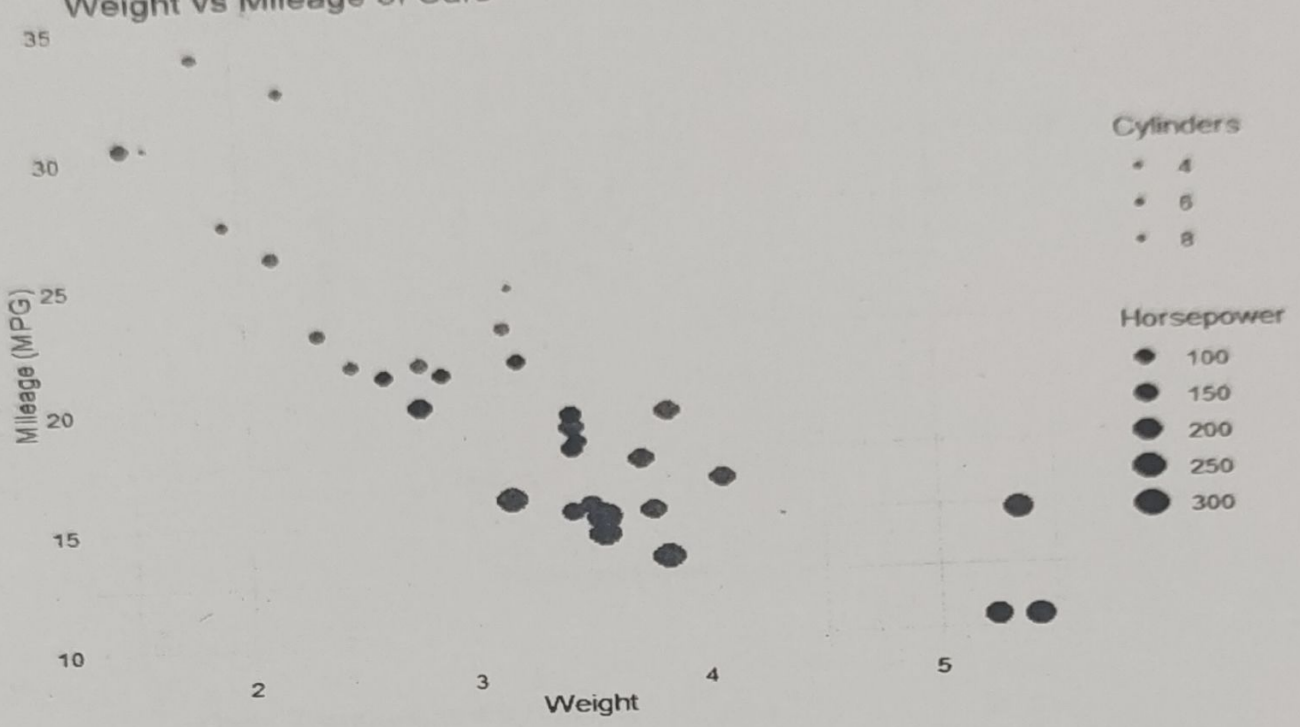
Assignment - II

Visualizing Amounts:

: To visualize and compare quantities

Q1. Mapping Data onto Aesthetics

a.

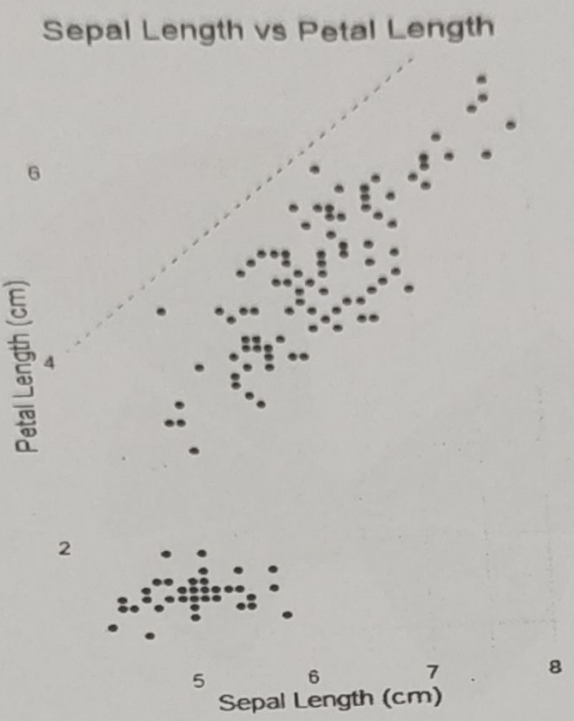


b.

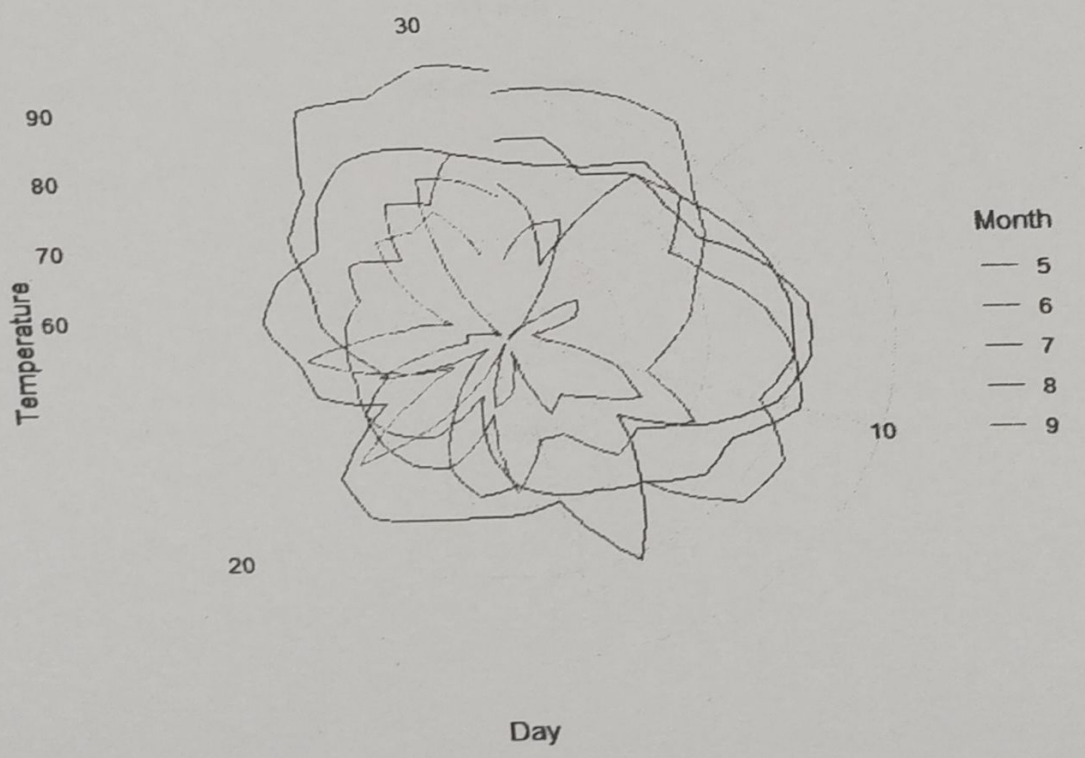
	Variable	Type	Suitable_Aesthetic
1	manufacturer	Qualitative	Color
2	model	Quantitative	Position
3	displ	Quantitative	Position
4	year	Quantitative	Size
5	cyl	Quantitative	Position
6	trans	Qualitative	Color
7	drv	Quantitative	Position
8	cty	Quantitative	Position
9	hwy	Qualitative	Color
10	fl	Qualitative	Shape
11	class	Qualitative	Shape

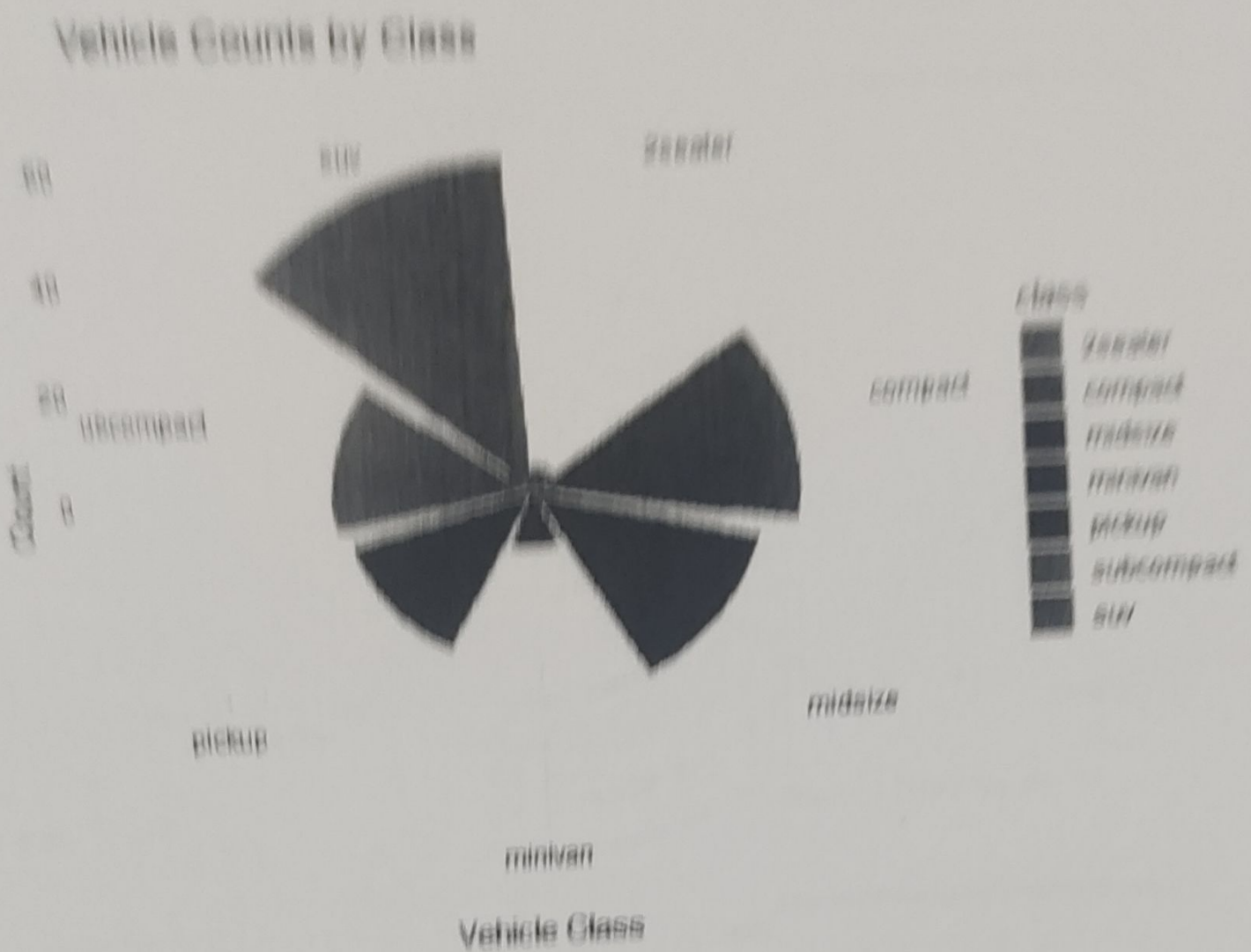
Q2. Coordinate Systems and Axes

a.



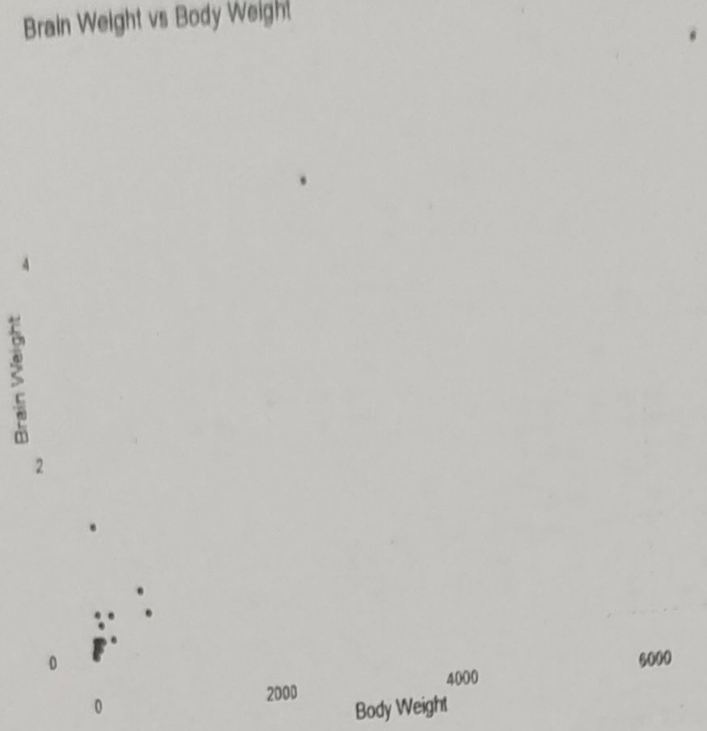
b.



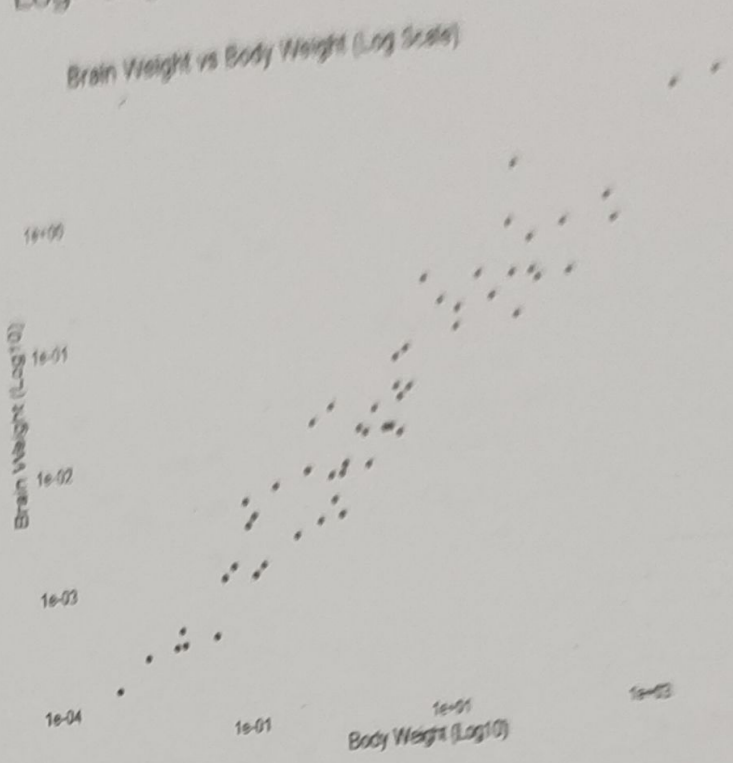

Vehicle Counts by Class
suv
2seater
60
40
20
0
Count
subcompact
compact
midsize
minivan
pickup
Class
2seater
compact
midsize
minivan
pickup
subcompact
suv
pickup
midsize
minivan
Vehicle Class

Q3. Nonlinear Axes

a. Linear Scale

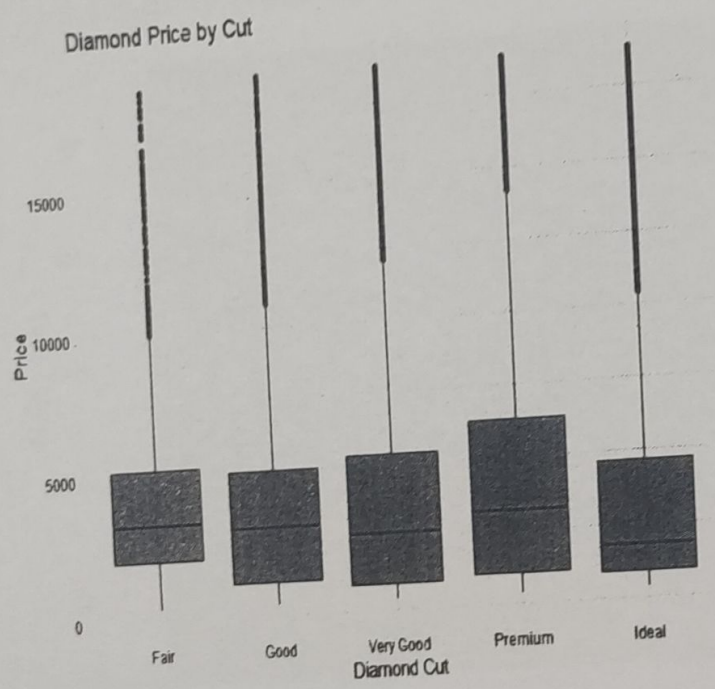


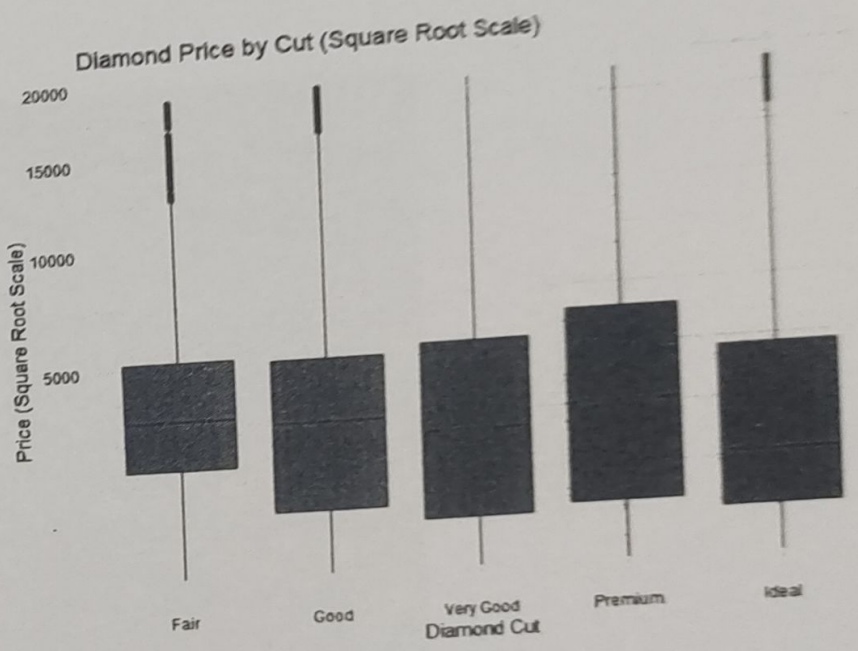
Log - Log Scale



Square Root Scale

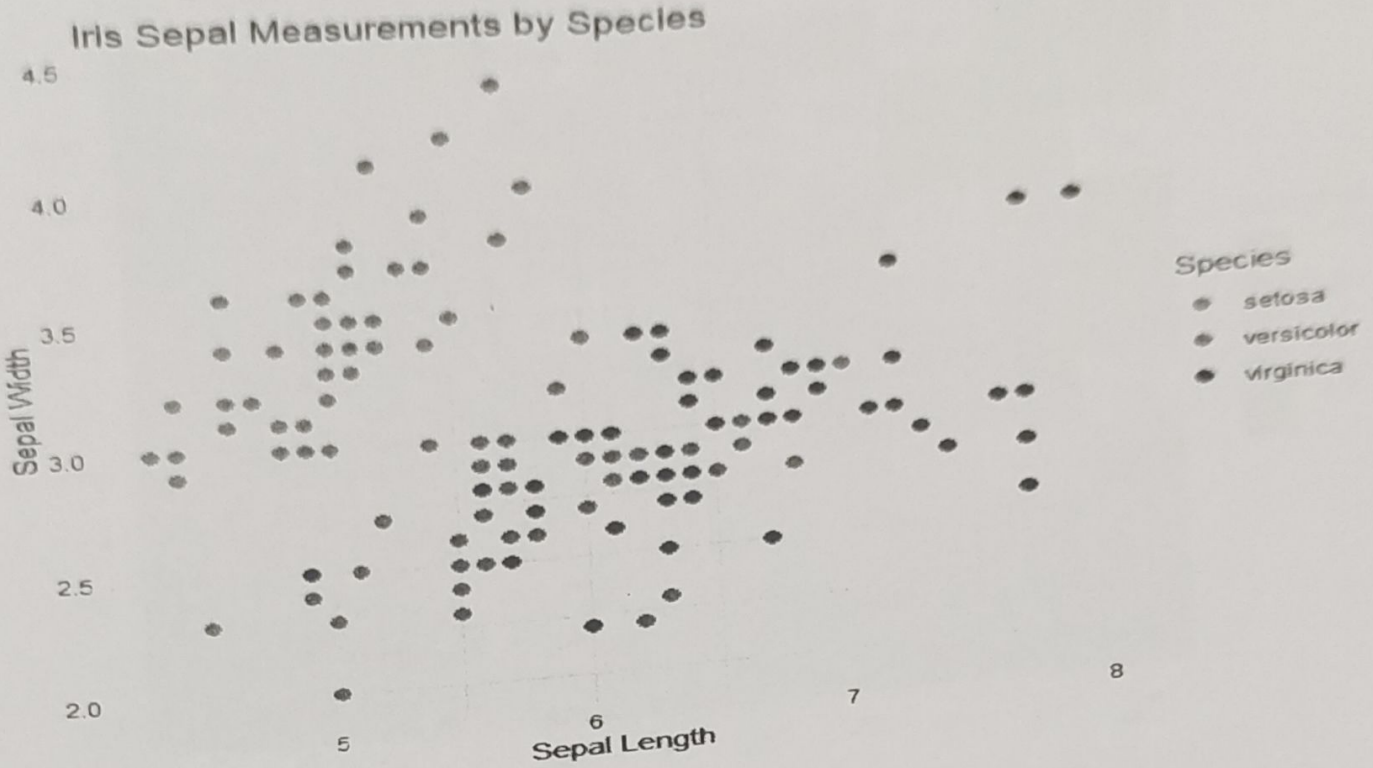
b. Linear Scale



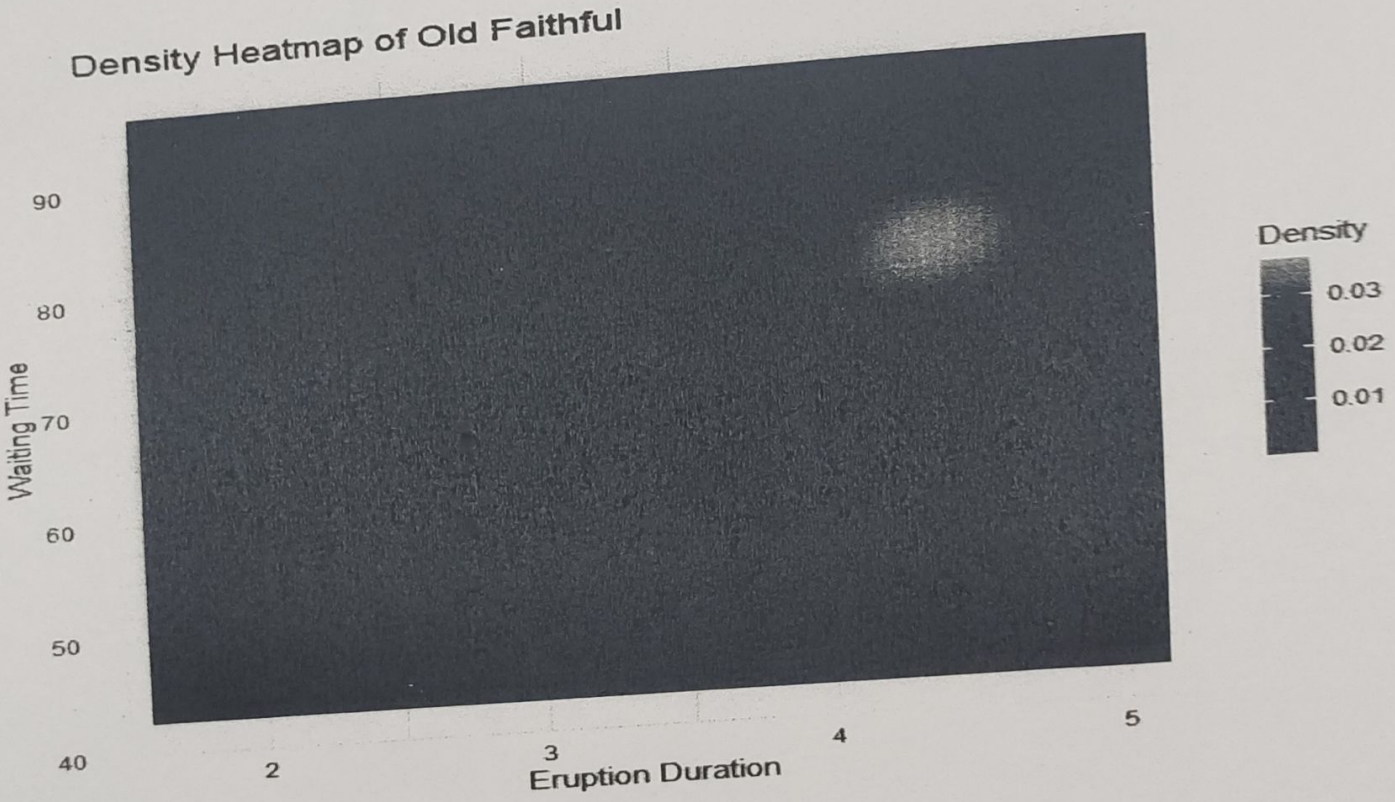


Q4. Color Scales: Qualitative, Sequential and Diverging

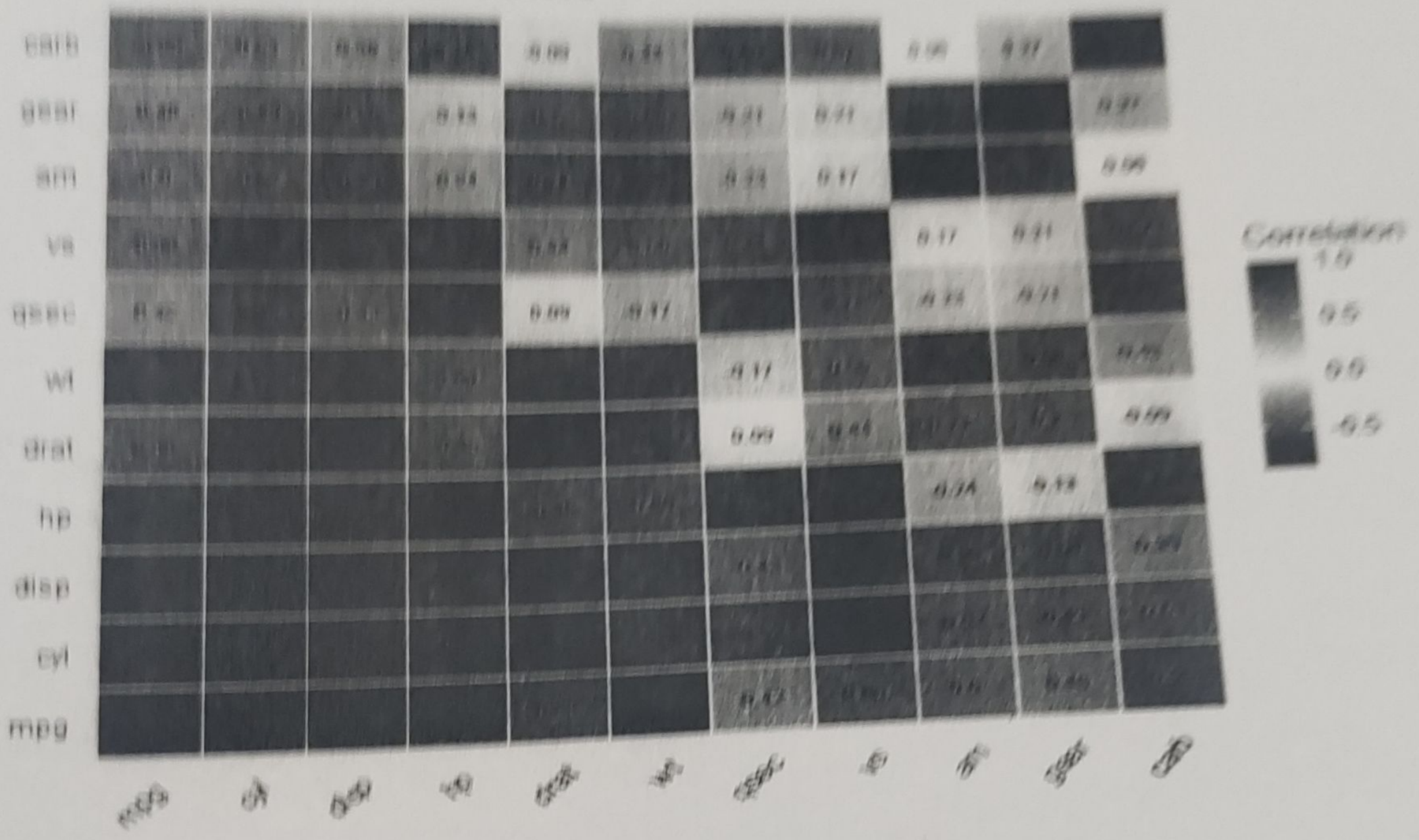
a.



b.

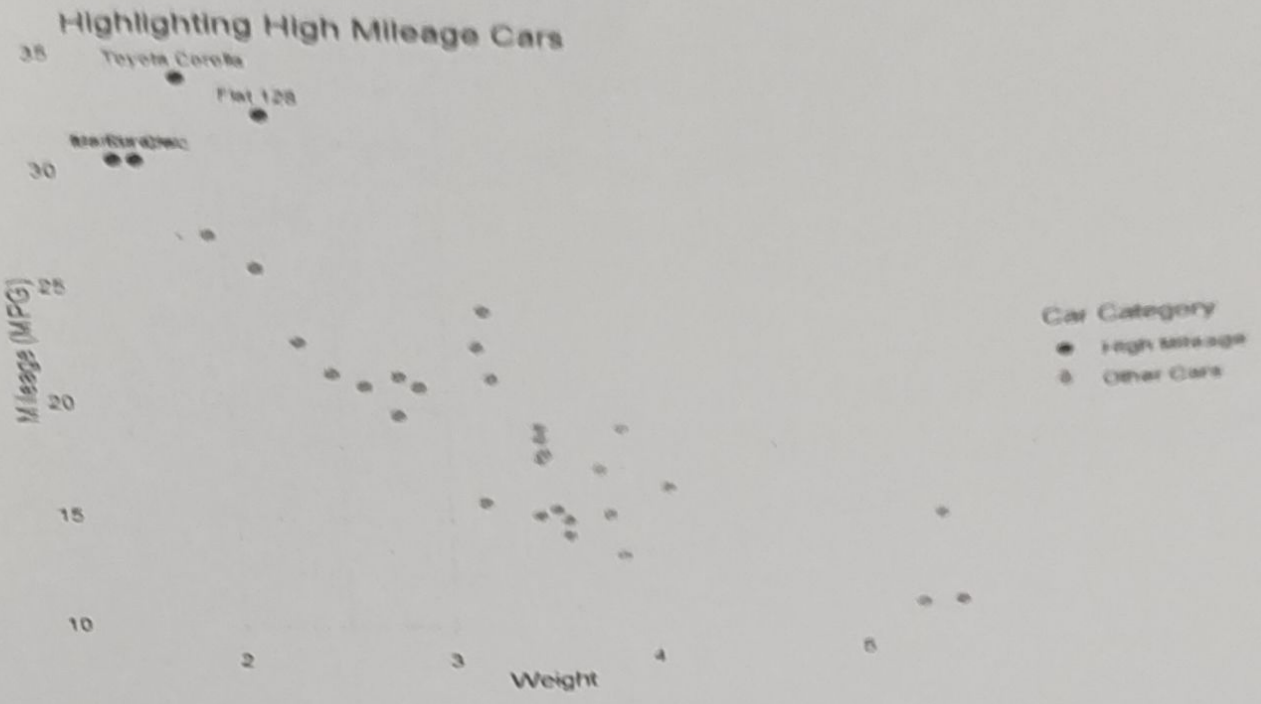


Correlation Matrix of mtcars



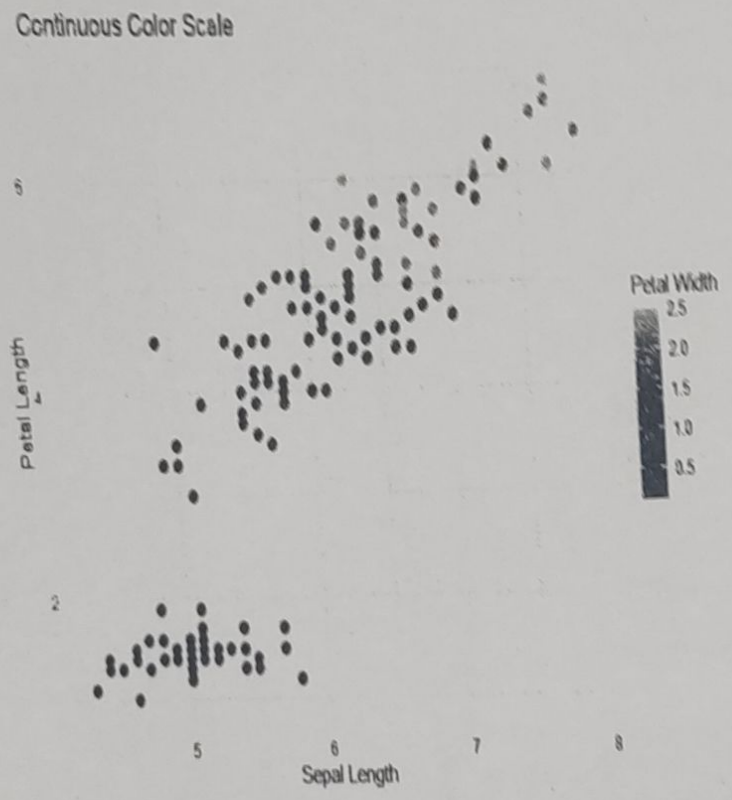
Q5. Color as a Tool to Highlight and Continuous vs Discrete Scales

a.



b.

Continuous Color



Discrete Color

